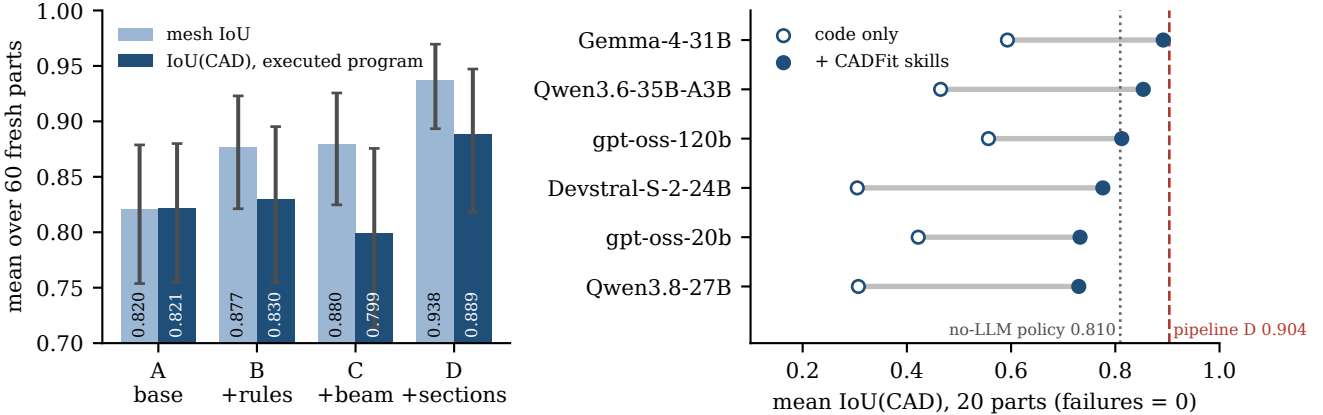
(a) Search arms: mesh vs executed CAD
mean over 60 fresh parts
1.00
0.95
0.90
0.85
0.80
0.75
0.70
mesh IoU
IoU(CAD), executed program
0.820
0.821
0.877
0.830
0.880
0.799
0.938
0.889
A
base
B
+rules
C
+beam
D
+sections
(b) Open-weight agents: code vs skills
code only
+ CADFit skills
Gemma-4-31B
Qwen3.6-35B-A3B
gpt-oss-120b
Devstral-S-2-24B
gpt-oss-20b
Qwen3.8-27B
no-LLM policy 0.810
pipeline D 0.904
0.2
0.4
0.6
0.8
1.0
mean IoU(CAD), 20 parts (failures = 0)